

Muhammad Ali Khan

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EDUCATION

University of Pécs

BSc Computer Science

Pécs, Hungary

Sep. 2023 – Present

Aalto University Summer School

Introduction to Artificial Intelligence and Machine Learning

Helsinki, Finland

July 2025 – Aug. 2025

Saxion University of Applied Sciences

BSc Applied Computer Science and Electronics

Enschede, Netherlands

Sep. 2022 – Aug. 2023

EXPERIENCE

Research Intern

Trustworthy Artificial Intelligence Laboratory (TAIL), University of Cagliari

May 2026 – Present

Cagliari, Italy

- Working on Retrieval-Augmented Generation (RAG) systems using the FlashRAG framework.
- Developing and evaluating retrieval pipelines, datasets and benchmarks.
- Implemented Mosaic-RAG microservice architecture using FastAPI and Docker.
- Conducting experiments on information retrieval and large language model systems.

AI & Machine Learning Summer School Participant

Aalto University

Jul. 2025 – Aug. 2025

Helsinki, Finland

- Completed intensive training in Artificial Intelligence and Machine Learning.
- Developed computer vision applications using YOLOv8.
- Worked on practical deep learning and data science projects.

PROJECTS

Traffic and Crowd Monitoring System [GitHub] | Python, OpenCV, YOLOv8, PostgreSQL

2026

- Developing a BSc thesis project for real-time traffic and crowd monitoring.
- Implemented object detection, counting and analytics using YOLOv8.
- Integrated database storage and visualization of monitoring data.

Mosaic-RAG [GitHub] | Python, FastAPI, Docker

2026

- Designed a modular Retrieval-Augmented Generation architecture.
- Built independent retrieval and generation microservices.
- Implemented REST APIs and Docker-based deployment.

Object Detection using YOLOv8 [GitHub] | Python, Computer Vision

2025

- Developed object detection models during the Aalto University Summer School.
- Applied deep learning techniques to image and video analysis tasks.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript

Artificial Intelligence: Retrieval-Augmented Generation (RAG), Computer Vision, Deep Learning, Large Language Models (LLMs)

AI & ML Frameworks: PyTorch, YOLOv8, OpenCV, ONNX Runtime, LangChain, FAISS

Backend Development: FastAPI, Flask, REST APIs, Microservices

Databases: PostgreSQL, MySQL

DevOps & Tools: Git, Docker, Linux, GitHub Actions, WSL2

Cloud: AWS (Solutions Architect Associate – In Progress)